

Brain-Computer Interface Rehabilitation System Design Based on Motor Imagery

Quanpeng Wei¹, Airong Wei¹, Fulai Peng², Xikun Zhang², Xingwei Wang², Zhiyong Wang², Peiran Li¹

1. School of Control Science and Engineering, Shandong University, Jinan, China

2. Shandong Institute of Advanced Technology, Chinese Academy of Sciences, Jinan, China

sduwqp@163.com, weairong@sdu.edu.cn, pengfulai1112@163.com,

xk.zhang@sdiat.ac.cn, xwwang051@163.com, w0017@hotmail.com, lipr971114@163.com

Corresponding Author: Airong Wei Email: weairong@sdu.edu.cn

Fulai Peng Email: pengfulai1112@163.com

Abstract—To improve the rehabilitation efficacy for stroke patient, a brain-computer interface (BCI) rehabilitation system based on motor imagery (MI) is investigated in this paper. Firstly, the Electroencephalogram (EEG) signals are preprocessed by Laplacian spatial filter and Butterworth band pass filter with cutoff frequency of 8Hz~24Hz to enhance the interesting components of the EEG signals. Then, the logarithmic band power (LBP) features are extracted and the motion intentions classification model is established based on linear discriminant analysis (LDA) algorithm. Finally, the correct motion intentions recognition results are converted to executive command of our designed rehabilitation hand. The motion imagery dataset from OpenVibe is used to validate our system. Results show that the rehabilitation hand is driven to effectively actuate the hand to perform the pre-set actions according to the MI recognition results.

Keywords—Brain-computer interface; Motor imagery; Rehabilitation system; Stroke

I. INTRODUCTION

Stroke, also known as cerebral apoplexy and cerebrovascular accident, is a kind of disease with high morbidity, disability, mortality and recurrence rate [1,2]. The traditional therapy approach is indirectly stimulating the nervous system corresponding to the patient's movement to achieve the therapeutic effect. However, all physical therapy methods require the participation of therapeutic physicians. In some developing countries, medical services cannot be provided to all patients due to the large number of stroke patients and the shortage of rehabilitation doctors.

With the development of science and technology, ever-growing scholars have investigated feasible methods in the treatment of stroke rehabilitation. Mainstream stroke rehabilitation therapy includes mirror therapy, robot intervention, virtual reality feedback training and transcranial direct current stimulation (tDCS) [3-6]. In [7], the researchers discuss various adaption options and their

implications in stroke rehabilitation using Electroencephalogram (EEG) based brain-computer (BCI) interface. In [8], the researchers applied meta-analysis to compare the effects of methods used in stroke rehabilitation, and derived the conclusion that BCI rehabilitation therapy had the better effect in promoting motor function recovery of stroke patients. The mental state used by BCI is kinesthetic imagination/attempt to move the target limb to promote the cortical reorganization of the damaged hemisphere [9,10]. BCI operated using power modulations related to inhibition of the contralesional side and excitation of the ipsilesional side have been shown to be effective for post-stroke motor recovery [11,12].

In this paper, a portable stroke rehabilitation training system based on motor imagery (MI) BCI is proposed. Firstly, Laplacian spatial filter and band pass filter with cutoff frequency of 8Hz~24Hz are used to preprocess the signals to enhance the interesting components from the EEG signals. Then, logarithmic band power (LBP) features are extracted to establish a motion intentions classification model based on linear discriminant analysis (LDA) algorithm. Finally, the correct motion intentions recognition results are translated into executive command of our designed rehabilitation hand, which will actuate the disabled hand to perform the pre-set actions.

II. DESIGN OF BRAIN COMPUTER INTERFACE REHABILITATION SYSTEM

The rehabilitation based on BCI consists of several key procedures: EEG data acquisition, signal preprocessing, feature extraction and classification construction and hand rehabilitation according to the motion classification results. The flowchart of the rehabilitation is presented in Fig. 1, which consists of two stages, the first one is the offline motor intentions classification model constructing stage and the second one is the online rehabilitation training stage. All algorithms of the signal processing and classification in this article are implemented in OpenVibe platform.

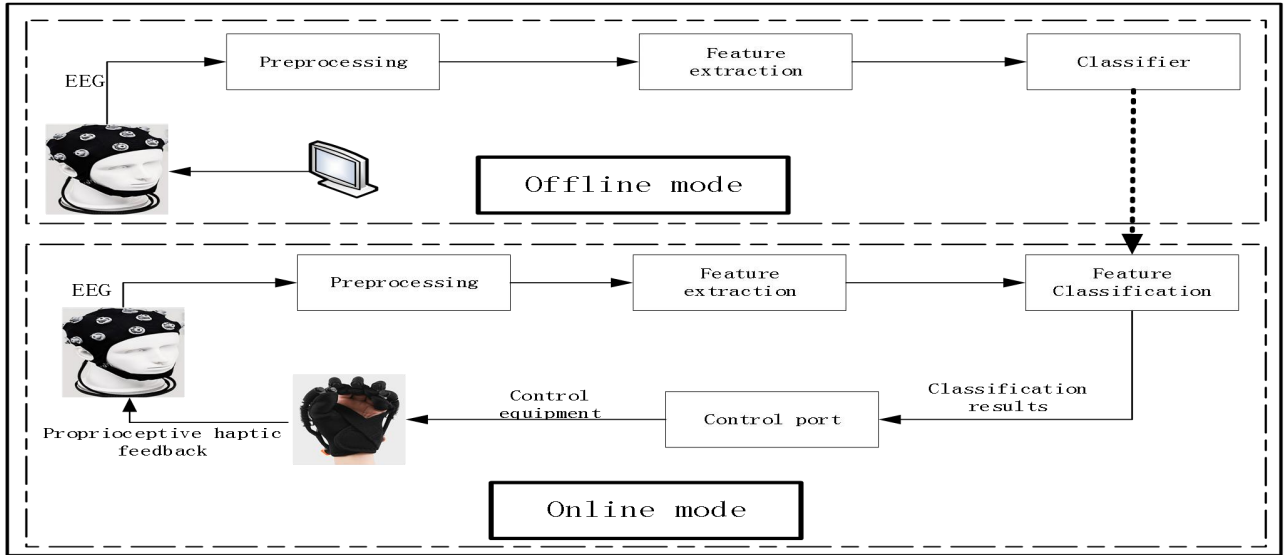


Fig. 1. Operation flow of the rehabilitation system.

A. EEG data acquisition

The EEG channels used in this paper are as follows: CPz (reference electrode), AFz (ground electrode), C3, C4, FC3, FC4, C5, C1, C2, C6, CP3, CP4. The EEG electrode placement is shown in Fig. 2.

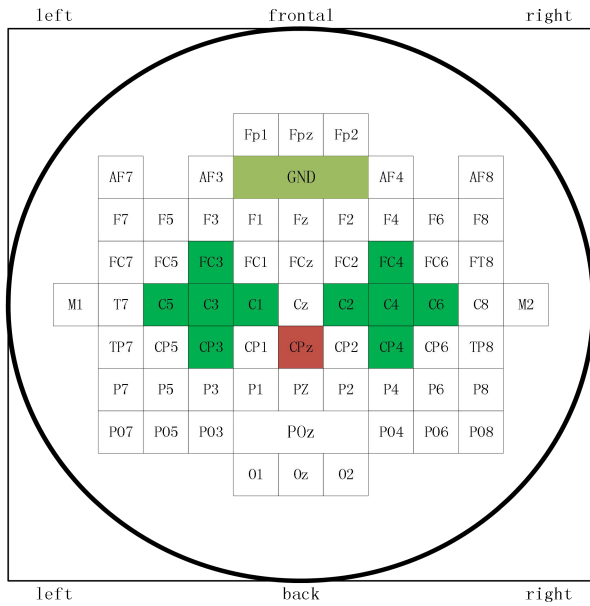


Fig. 2. The EEG electrode placement

The data acquisition process consists of two stages, model training stage and online rehabilitation stage. In the training stage, the subject performed two times imagination tasks according to the left/right arrow clue. Then the motor recognition model is constructed. In the online rehabilitation stage, the subject also performed the left/right hand movement imagination according to the arrow clue. The rehabilitation hand will actuate the hand to perform grasp actions when the motion recognition

results is correct. In addition, the blue bar on the display extending to the corresponding directions serves as visual feedback. Each trial persists 8 seconds: in the first 2 seconds the subject is relaxed, at the 2nd second, the cross clue appears to remind the subject to start the experiment, then at the 3rd second, the left/right arrow clue appears, the subject is instructed to start imagining movements of the left or right hand according to the arrow clue. The subject need to continue imagining the hand movement last for four seconds. In the last one second, the corresponding direction blue bar appears as visual feedback.

The time course of a single trial is shown in Fig. 3. Both the model training stage and online rehabilitation stage perform the same trial process.

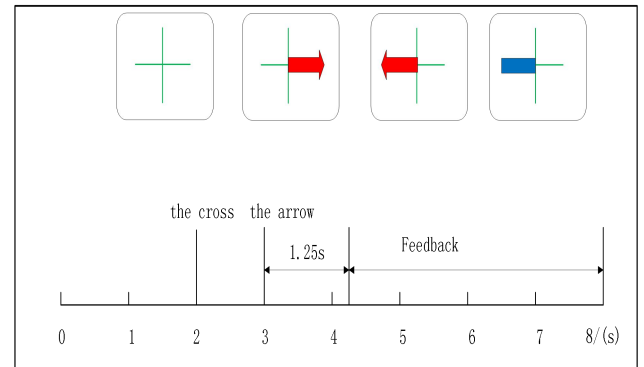


Fig. 3. Time course of a single trial.

B. Signal preprocessing

The 10 channel EEG signals are firstly re-referenced by the laplacian filter to reduce the effect of common components of neighboring signals. Considering that C3 and C4 location are the main regions mirroring the

left/right hand movements, we re-reference the C3 and C4 signals by the channels surrounding them respectively. More specifically, the C3 channel is re-referenced by C5, FC3, C1, CP3, and the C4 channel is re-referenced by C2, FC4, C6, CP4. The re-referenced output channels are computed as

$$OC_k = \text{Sum on } j(S_{jk} * IC_j).$$

where IC_j is the j_{th} input channel, OC_k is the k_{th} output channel, and S_{jk} is the coefficient for the j_{th} input channel and k_{th} output channel in the Spatial filter matrix.

In this paper, the input EEG-channels are: C3, C4, FC3, FC4, C5, C1, C2, C6, CP3, CP4.

The spatial filter coefficients are: 4, 0, -1, 0, -1, -1, 0, 0, -1, 0, 0, 4, 0, -1, 0, 0, -1, -1, 0, -1.

The number of output channels are two and the number of input channels are ten.

Specifically, the output channels of spatial filter are as follows,

The number of output channels are two and the number of input channels are ten.

Specifically, the output channels of spatial filter are

$$\begin{aligned} OC1 &= 4 * C3 + 0 * C4 + (-1) * FC3 + 0 * FC4 + (-1) * C5 \\ &\quad + (-1) * C1 + 0 * C2 + 0 * C6 + (-1) * CP3 + 0 * CP4 \\ &= 4 * C3 - FC3 - C5 - C1 - CP3 \\ OC2 &= 0 * C3 + 4 * C4 + 0 * FC3 + (-1) * FC4 + 0 * C5 \\ &\quad + 0 * C1 + (-1) * C2 + (-1) * C6 + 0 * CP3 + (-1) * CP4 \\ &= 4 * C4 - FC4 - C2 - C6 - CP4 \end{aligned}$$

where $OC1$ and $OC2$ represent the re-referenced C3 and C4, respectively.

The two channel output signals are then filtered by Butterworth band pass filter with cutoff frequency of 8Hz~24Hz to remove the uninterested components.

C. Feature Extraction

After filter processing, the features that mostly represent the MI information are then extracted. In this paper, we applied the LBP algorithm to extract the features due to its low computational complexity. The LBP feature is computed as follows

$$P_i = \frac{1}{N} \sum_{j=1}^N a_j^2 \quad (i = 1, 2)$$

where a_j is the j_{th} signal sampling point, P_i is the mean band power.

The LBP is then calculated as

$$F_i = \log(P_i + 1) \quad (i = 1, 2).$$

D. Classification model construction

In this part, we use the LDA algorithm to construct the classification model. The theory of LDA algorithm is deduced below.

Assuming that the EEG data set is $D = \{x^{(1)}, x^{(2)}, \dots, x^{(n)}\}$. D have two subsets D_1 and D_2 , satisfying $D = D_1 \cup D_2$.

The data in D_1 belongs to classes C_1 , and the data in D_2 belongs to classes C_2 . We get the average of the two

$$\text{subsets } \mu_1 = \frac{1}{n_1} \sum_{x^{(i)} \in D_1} x^{(i)}, \mu_2 = \frac{1}{n_2} \sum_{x^{(i)} \in D_2} x^{(i)}.$$

Since there are two types of data, we only need to project the data onto a line. For a given vector w , we

$$\text{define } y^{(i)} = w^T x^{(i)}, \tilde{\mu}_1 = \frac{1}{n_1} \sum_{y^{(i)} \in D_1} y^{(i)}, \tilde{\mu}_2 = \frac{1}{n_2} \sum_{y^{(i)} \in D_2} y^{(i)}.$$

Clearly, the following relationships exist

$$\tilde{\mu}_1 = w^T \mu_1, \tilde{\mu}_2 = w^T \mu_2.$$

$$|\tilde{\mu}_1 - \tilde{\mu}_2|^2 = w^T (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T w.$$

Define standard deviations σ_1 and σ_2 , where

$$\sigma_1^2 = \sum_{y^{(i)} \in D_1} (y^{(i)} - \tilde{\mu}_1)^2, \sigma_2^2 = \sum_{y^{(i)} \in D_2} (y^{(i)} - \tilde{\mu}_2)^2.$$

Convert the above two variance formulas as follows

$$\begin{aligned} \sigma_1^2 &= \sum_{y^{(i)} \in D_1} (y^{(i)} - \tilde{\mu}_1)^2 \\ &= \sum_{x^{(i)} \in D_1} (w^T x^{(i)} - w^T \mu_1)^2 \\ &= w^T S_1 w \\ \sigma_2^2 &= \sum_{y^{(i)} \in D_2} (y^{(i)} - \tilde{\mu}_2)^2 \\ &= \sum_{x^{(i)} \in D_2} (w^T x^{(i)} - w^T \mu_2)^2 \\ &= w^T S_2 w \end{aligned}$$

where

$$\begin{aligned} S_1 &= \sum_{x^{(i)} \in D_1} (x^{(i)} - \mu_1)(x^{(i)} - \mu_1)^T \\ S_2 &= \sum_{x^{(i)} \in D_2} (x^{(i)} - \mu_2)(x^{(i)} - \mu_2)^T. \end{aligned}$$

LDA requires that the center distance between different types of data is as large as possible, and the projection points of data of the same category to be as close as possible. Therefore, we should maximize $|\tilde{\mu}_1 - \tilde{\mu}_2|^2 = w^T (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T w$, and make the covariance σ_1^2 and σ_2^2 of the projection points of the same sample as small as possible. In summary, our optimization objectives are

$$\begin{aligned} \arg \max_w J(w) &= \frac{\|w^T \mu_1 - w^T \mu_2\|_2^2}{w^T S_1 w + w^T S_2 w} \\ &= \frac{w^T (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T w}{w^T (S_1 + S_2) w} \end{aligned}$$

Define the intra-class dispersion matrix

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T.$$

Define the interclass dispersion matrix

$$S_w = S_1 + S_2.$$

In this way, the optimization goal is rewritten as

$$\arg \max_w J(w) = \frac{w^T S_B w}{w^T S_w w}.$$

Differentiate the above formula, we can obtain

$$w = \lambda S_w^{-1} w.$$

At the same time, $S_B w$ and $\mu_1 - \mu_2$ are parallel. Consequently, we just need to find w which makes $S_w w$ and $\mu_1 - \mu_2$ parallel. If S_w is nonsingular, then we can take $S_B w = S_w^{-1} (\mu_1 - \mu_2)$.

After solving a w , we get a binary training model and the formula is

$$\arg_{i \in \{1,2\}} (w^T x - \mu_i).$$

E. Design of rehabilitation hand

The motion intentions recognized by the above procedures are then converted to executive command of the rehabilitation hand, which will actuate the disabled hand to perform the pre-set actions. The commands are transmitted to the rehabilitation hand through the serial port.

Rehabilitation hand equipment is composed of Arduino UNO R3 microcontroller, four-way optical coupling isolation relay driver, air pump, two-position three-way valve FA2021D and normally closed solenoid valve FA2021C. The circuit diagram of the rehabilitation hand is shown in Fig. 4.

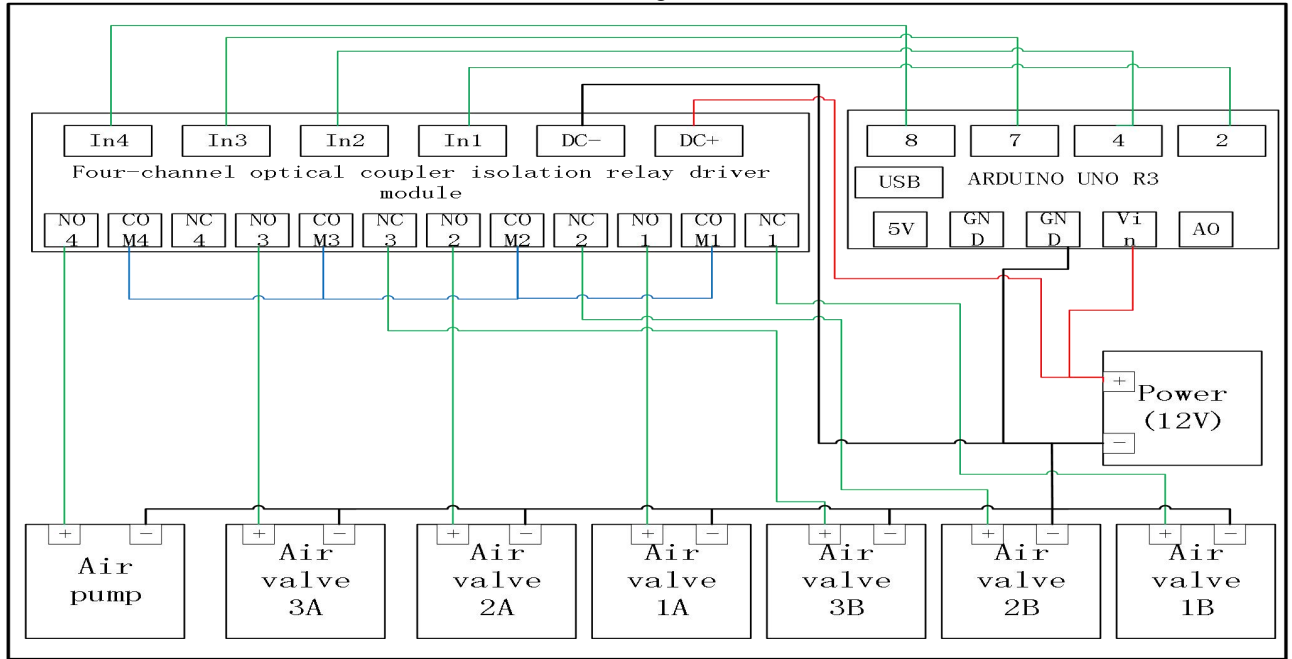


Fig. 4. Operation flow of the rehabilitation system.

The rehabilitation hand controls the switch of the isolation relay through the single chip microcomputer. The state of the relay determines whether the pump and valve can be energized. The air pump and the air valve cooperate with each other to determine the inflation and

ventilation of the rehabilitation gloves. All four modules of the relay are set as high-level trigger. The corresponding relationship among relay, air valve, air pump and the state of rehabilitated hand is shown as Table 1.

TABLE I. THE CORRESPONDING RELATIONSHIP BETWEEN RELAY, AIR VALVE, AIR PUMP AND THE STATE OF REHABILITATED HAND

<i>Level value</i> (In1,In2,In3,In4)	<i>Air pump</i>	<i>Air valve 1A</i>	<i>Air valve 2A</i>	<i>Air valve 3A</i>	<i>Air valve 1B</i>	<i>Air valve 2B</i>	<i>Air valve 2B</i>	<i>The state of rehabilitation hand</i>
(LOW,LOW,LOW,LOW)	power off	close	close	close	open	open	open	Not work
(HIGH,HIGH,HIGH,HIGH)	power on	open	open	open	close	close	close	Left glove inflates
(HIGH,LOW,LOW,LOW)	power off	open	close	close	close	open	open	Left glove deflates
(LOW,LOW,LOW,HIGH)	power on	close	close	close	open	open	open	Right glove inflates
(LOW,HIGH,HIGH,LOW)	power off	close	open	open	open	close	close	Right glove deflates

III. RESULTS

To evaluate the efficacy of the proposed system, in this paper, we only used the dataset provided by OpenVibe to perform the test. There are 36 times of movement imagination which include 19 trials of left hand and 17 trials of right hand in the OpenVibe's dataset. The EEG signals are captured from motor cortex regions (Nz (reference), C3, C4, FC3, FC4, C5, C1, C2, C6, CP3 and CP4 channels). The recognition results on OpenVibe's dataset are as follows: training accuracy rate is 80.05%, and the accuracy rate of 7-fold cross-validation is 77.49% in the test set. During the test phase, the movement of the rehabilitated hand can be well controlled based on the recognition results.

IV. CONCLUSIONS

In this paper, we have proposed a MI-based rehabilitation training system for stroke patient, which consists of the following key procedures. Firstly, the signals are preprocessed by Laplacian spatial filter and band pass filter with cutoff frequency of 8Hz~24Hz to enhance the interesting components from the EEG signals. Then, the LBP features that represent the motion information are extracted and the motion intentions classification model is established based on LDA algorithm. Finally, the correct motion intentions recognition results are converted to executive command of our designed rehabilitation hand, which actuates the disabled hand to perform the pre-set actions. The rehabilitation hand can effectively actuate the hand to perform the pre-set actions according to the recognition results.

ACKNOWLEDGMENT

This research was supported by the National Nature Science Foundation of China (61973188), the project of Jinan 20 Universities (2019GXRC040), Natural Science Foundation of Shandong Province (ZR2020QF024), Jinan 5150 Program for Talents Introduction and Shandong Institute of Advanced Technology, Chinese Academy of Sciences (YJZX003).

REFERENCES

- [1] Campbell, Bruce C V et al. Endovascular therapy for ischemic stroke with perfusion-imaging selection." The New England journal of medicine vol. 372,11 (2015): 1009-18.
- [2] Saver, Jeffrey L et al. "Stent-retriever thrombectomy after intravenous t-PA vs. t-PA alone in stroke." The New England journal of medicine vol. 372,24 (2015): 2285-95.
- [3] G. Salhab, A. R. Sarraj and S. Saleh, "Mirror therapy combined with functional electrical stimulation for rehabilitation of stroke survivors' ankle dorsiflexion," 2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2016, pp. 4699-4702.
- [4] Y. Ren, Y. -N. Wu, C. -Y. Yang, T. Xu, R. L. Harvey and L. -Q. Zhang, "Developing a Wearable Ankle Rehabilitation Robotic Device for in-Bed Acute Stroke Rehabilitation," in IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 25, no. 6, pp. 589-596, June 2017.
- [5] S. I. Kim et al., "Proprioception rehabilitation training system for stroke patients using virtual reality technology," 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2013, pp. 4621-4624.
- [6] H. Yu et al., "Effects of Transcranial Direct Current Stimulation on Stroke Based on Brain Functional Networks," in IEEE Transactions on Magnetics, vol. 57, no. 6, pp. 1-4, June 2021, Art no. 5800404.
- [7] C. Guan, "Adaptation in motor imagery brain-computer interfaces and its implication in rehabilitation," 2016 4th International Winter Conference on Brain-Computer Interface (BCI), 2016, pp. 1-2.
- [8] Cervera MA, Soekadar SR, Ushiba J, Millán JDR, Liu M, Birbaumer N, Garipelli G. Brain-computer interfaces for post-stroke motor rehabilitation: a meta-analysis. Ann Clin Transl Neurol. 2018 Mar 25;5(5):651-663.
- [9] Dobkin, B.H. (2007), Brain-computer interface technology as a tool to augment plasticity and outcomes for neurological rehabilitation. The Journal of Physiology, 579: 637-642.
- [10] Dodd KC, Nair VA, Prabhakaran V. Role of the Contralesional vs. Ipsilesional Hemisphere in Stroke Recovery. Front Hum Neurosci. 2017 Sep 21;11:469.
- [11] Buch E, Weber C, Cohen LG, Braun C, Dimyan MA, Ard T, Mellinger J, Caria A, Soekadar S, Fourkas A, Birbaumer N. Think to move: a neuromagnetic brain-computer interface (BCI) system for chronic stroke. Stroke. 2008 Mar;39(3):910-7.
- [12] Coscia M, Wessel MJ, Chaudary U, Millán JDR, Micera S, Guggisberg A, Vuadens P, Donoghue J, Birbaumer N, Hummel FC. Neurotechnology-aided interventions for upper limb motor rehabilitation in severe chronic stroke. Brain. 2019 Aug 1;142(8):2182-2197.